

A top-down view of a light teal desk. In the top left is a silver laptop with a black keyboard. To its right is a white mouse and a dark coffee cup. In the bottom left is a brown leather notebook with a white pen and a pair of tortoiseshell glasses.

# AIClub





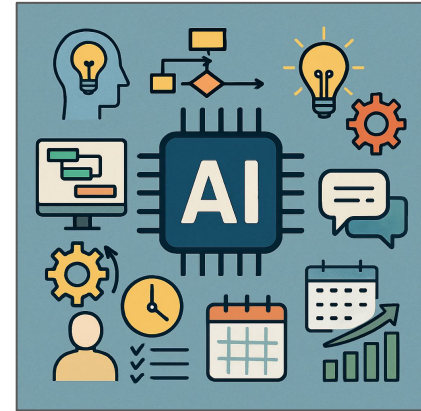
# AI Project Cycle

# What is AI Project Cycle?

AI Project Cycle is a step-by-step process used to build an AI-based solution.

It helps us:

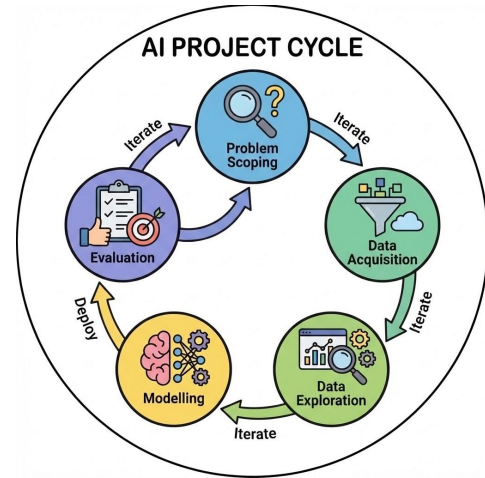
- Solve real-world problems
- Use data effectively
- Build intelligent systems



# The 5 Stages of AI Project Cycle

AI Project Cycle has 5 stages:

1. Problem Scoping
2. Data Acquisition
3. Data Exploration
4. Modelling
5. Evaluation

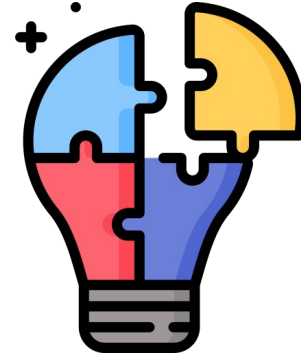


Each stage is connected and important.

# Stage 1 – Problem Scoping

Problem Scoping means clearly defining the problem.  
Before building AI, we must understand:

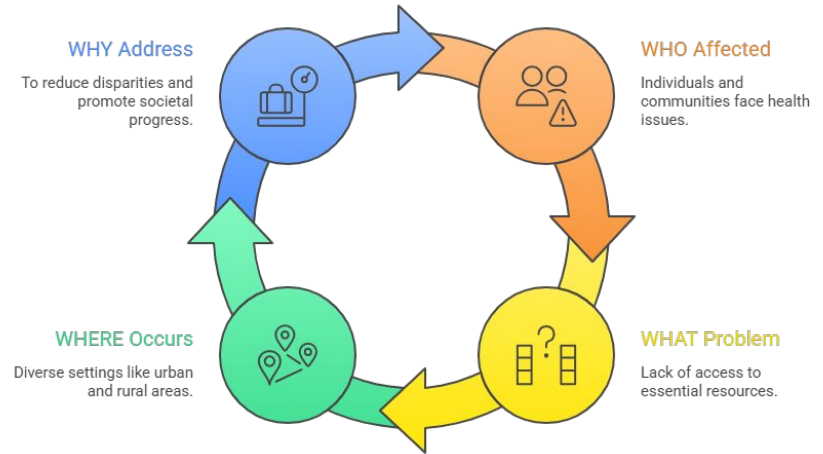
- What is the problem?
- Who is affected?
- Why is it important?



# 4Ws Framework

- **WHO** – Who is affected?
- **WHAT** – What is the problem?
- **WHERE** – Where does it occur?
- **WHY** – Why is it important?

A clear problem leads to a better solution.



## Stage 2 – Data Acquisition

Data Acquisition means collecting relevant data related to the problem.

- AI systems learn from data.
- Without data, AI cannot work.



# Types & Sources of Data

## Types:

- Structured (tables, numbers)
- Unstructured (images, videos, text)
- Sources:
  - Surveys
  - Sensors
  - Cameras
  - Websites
  - Databases





# Stage 3 – Data Exploration

Data Exploration means analyzing the collected data to find patterns and trends.

We check:

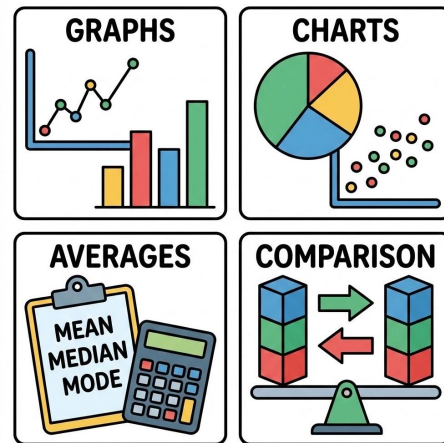
- Missing values
- Error
- Trends
- Relationships



# Tools Used in Data Exploration

- Graphs
- Charts
- Averages
- Comparison.

Goal: Understand data before building model.

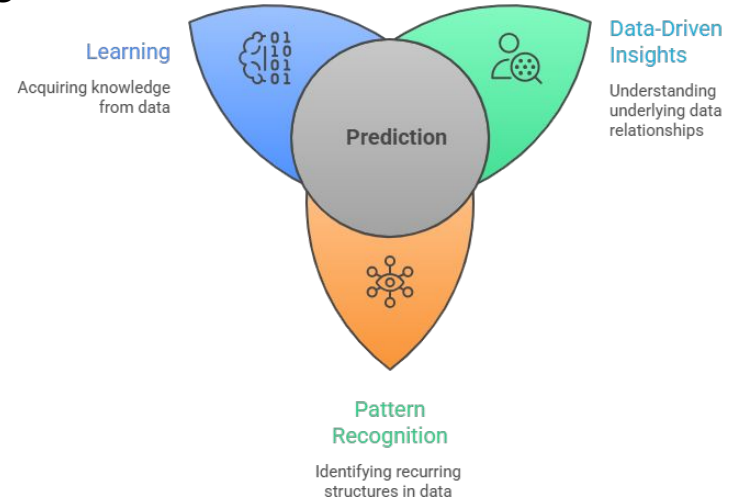


# Stage 4 – Modelling

Modelling means building an AI model using Machine Learning.

The model:

- Learns from data
- Finds patterns
- Makes predictions



# Training & Testing

## Training:

- AI learns from past data.

## Testing:

- AI is tested using new data.
- If performance is poor, improve model.

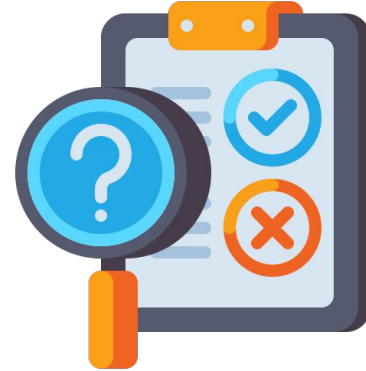


# Stage 5 – Evaluation

Evaluation means checking whether the AI model works correctly.

We measure:

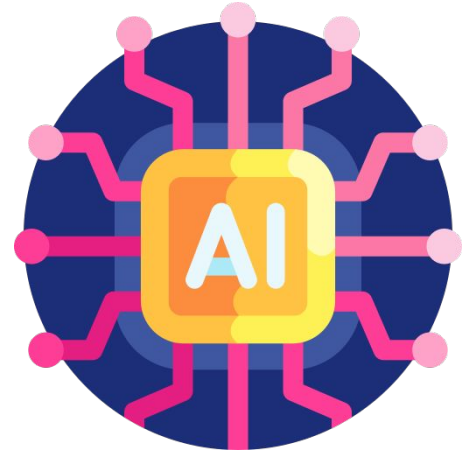
- Accuracy
- Efficiency
- Reliability



# Improving the Model

If model is not accurate:

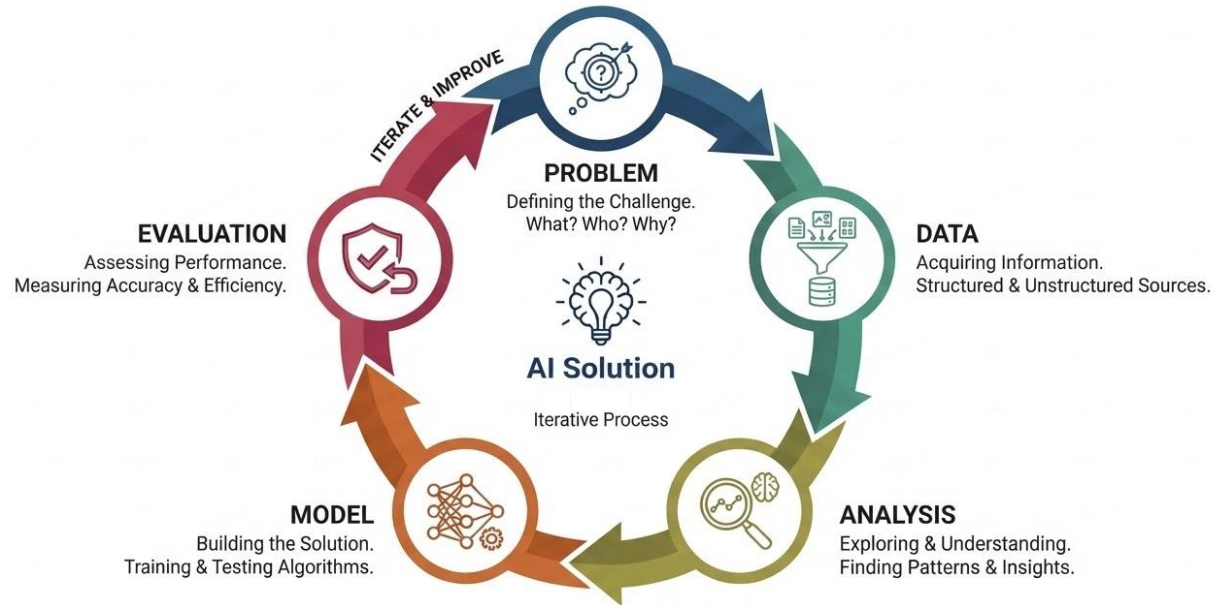
- Collect more data
- Improve data quality
- Change algorithm



AI Project Cycle is iterative (repeats until improvement).

# Complete Flow Diagram

## AI Project Cycle: From Problem to Solution



# Key Takeaways

- ✓ AI Project Cycle has 5 stages
- ✓ Each stage is important
- ✓ Data is central to AI
- ✓ The process is iterative



A top-down view of a desk with a light blue background. In the top left, a silver laptop is partially visible. Next to it is a white mouse. In the bottom left, there is a dark brown leather notebook with a white pen and a pair of tortoiseshell glasses.

# THANK YOU



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<https://aiclub.world>